

AvatarMirror: Rendering Restoration for Real-time Immersive Presence System

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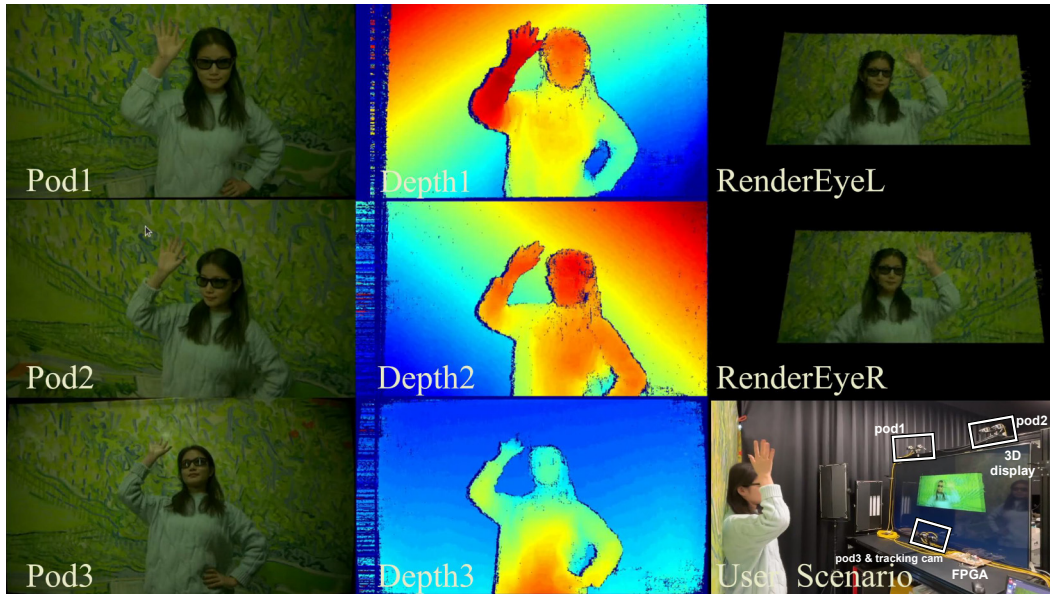


Figure 1: AvatarMirror: our real-time immersive presence system with instant novel-view 3D reconstruction and neural rendering. The user can observe himself with 3D optical glasses in novel views based on our system. Note that there are surrounding black areas because of the FOV (Field of View) gap of the imaging planes between the pods and the 3D display.

ABSTRACT

Human-centered immersive computing is catching on within VR and AR. We establish a real-time immersive presence system called AvatarMirror that enables arbitrary characters to look at themselves instantly in novel stereo views. First, stereo matching is implemented to acquire multi-view depth priors, based on which the human topology is reconstructed through raycasting from a novel perspective. Then, the color information from the referenced views is adaptively blended to perform the human rendering. Furthermore, we propose an RR-Net to conduct rendering restoration. AvatarMirror realizes real-time high-fidelity immersive presence with multi-threaded scheduling and CUDA acceleration.

Index Terms: Human-centered computing, Immersive display, Rendering, Raycasting, Image restoration, GAN.

1 INTRODUCTION

Displaying 3D digital humans on stereoscopic display devices is a challenging task. Recently, 3D Gaussian Splatting [3] has greatly improved novel-view rendering efficiency and quality. Several human-synthesis works [1] successfully leverage 3D Gaussian.

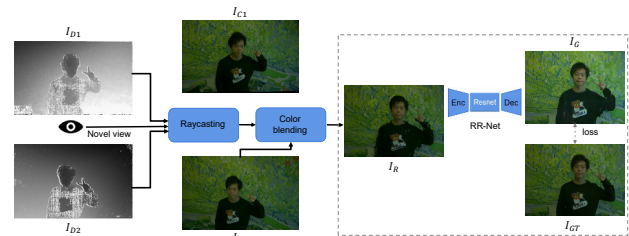


Figure 2: Pipeline of RR-Net where I_R and I_G are the results of 2.4.1 and 2.4.2, respectively. I_{GT} is from a predefined viewpoint for training. We take RGBD images from 2 pods as an example.

However, they need to learn human priors from a large amount of 3D scan data. We propose another efficient and effective approach with instant reconstruction and rendering to realize immersive human display. To achieve high-fidelity, real-time, and generalizable 3D digital human syntheses, we leverage multi-view depth priors and their corresponding aligned color images to reconstruct the topology of the digital human from a novel view and accurately render the colors for L&R eyes, which assists in the accurate presentation of a 3D digital human on a stereoscopic display device.

2 FRAMEWORK

As shown in Fig.1, based on the hardware, our proposed AvatarMirror framework mainly includes 4 modules: device calibration and synchronization, CUDA-based stereo matching, raycasting based on depth prior, and rendering with neural color restoration. The last two modules are based on new perspectives from eye tracking.

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2.1 Device calibration and synchronization

Device calibration includes camera calibration within the pod, extrinsic calibration between multiple pods, and calibration between the camera and the display screen. We apply Zhang’s [5] calibration method with a mirror and control the reprojection error within 0.2 pixels. We achieve multi-camera synchronization based on FPGA hardware triggering with a sampling frequency of 25Hz, and we synchronize camera captures by aligning timestamps.

2.2 CUDA-based stereo matching

After image rectification, we use Semi-Global Matching [2] for disparity search with CUDA acceleration. Then, depth is converted by means of $(focal_length * baseline) / |disparity + (c_{x1} - c_{x0})|$. We obtain synchronized RGBD information from 3 camera pods.

2.3 Raycasting based on depth prior

It is efficient to identify the topology for a digital human from a new perspective based on the depth priors of the reference views of 3 fixed pods. The CUDA-based algorithm procedure is as follows: a). Transform the camera coordinate of each pixel in the depth map to the corresponding position on a novel view. b). Locate the associated pixel coordinate onto the image plane from this new perspective. c). Store the depth information in the look-up table of the correlative pixel. d). Use 2D min/max filters to find the minimum and maximum depth values for each pixel on the novel view. e). We directly treat the minimum depth as the z-direction position of the corresponding point cloud.

2.4 Rendering with neural color restoration

2.4.1 Rendering based on depth prior

After localizing the human topology from the novel view, color fusion is conducted based on the color information from multiple pods. Specifically, given a point P that is visible in the new view, it is necessary to compare the z-direction position of P in the color camera’s perspective (denoted as P_c) with the precomputed depth prior for the same camera. If the depth value is less than the z-value of P_c , this indicates that P_c is occluded by another surface. In this case, the color blending weight for P_c should be set to 0, which does not aid in the rendering process of P . Otherwise, we apply equal weights to the relevant colors from different color maps.

2.4.2 Rendering restoration network

Due to the matching noise and holes in the depth prior generated by 2.2, the results of raycasting from a new perspective suffer from distortion and poor continuity, leading to subpar rendering quality (I_R in Fig 2). To address this issue, we propose RR-Net, which uses a lightweight deep learning model (9M parameters) to perform rendering restoration. The architecture specifically includes 3 down-sampling encoding layers, 3 ResNet layers, and 3 up-sampling decoding layers. The specific loss function is as follows:

$$L_{RR-Net} = L_{adv} + L_{rec} + \lambda_{vgg} L_{vgg} + \lambda_{FM} L_{FM} \quad (1)$$

, where L_{adv} tries to produce realistic images with adversarial learning, L_{rec} means the L1 regularization, L_{vgg} is the perceptual loss, L_{FM} means feature matching loss based on extracted discriminator features [4]. $\lambda_{vgg} = \lambda_{FM} = 10$. We train RR-Net with data augmentations, e.g., random flipping and cropping, and test on images of arbitrary sizes due to the fully convolutional structure.

3 EXPERIMENT

Implementation details. We use Basler aca1920-155uc as color camera with 1920*1080 resolution. We utilize KONKA LED65X9800U as the 3D display. The parameters for SGM are as follows: P1=10, P2=120. We set the cost aggregation path as 4 and the block size as 3. We employ a multi-threaded approach to control and process multiple tasks in parallel, including camera

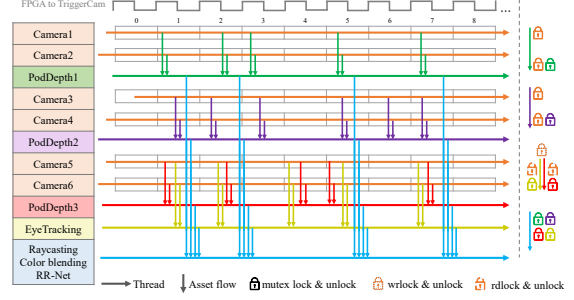


Figure 3: Visualization of our multi-threaded dynamic scheduling with mutex locks and write (wr) & read (rd) locks.

Table 1: Time study of our AvatarMirror framework. We achieve around 20 FPS on NVIDIA RTX 4080.

Module	SGM (CUDA)	Raycasting (CUDA) + Color blend (CUDA)	RR-Net (TRT)	Eye tracking (CPU)	Total
Time (ms)	15	6	8	30	59

capture, depth estimation, eye tracking, raycasting, color blending, and color restoration, as shown in Fig. 3. Independent control with mutexes ensures that these tasks are executed in a synchronized and orderly manner. Pod3 is also used to track eyes based on Dlib’s facial landmark detection. To simulate mirror-based observation, we reflect eye-tracking results to the behind of the 3D display. To ensure spatiotemporal consistency, we use the rendering result from the previous frame as the initial color and update the colors based on our rendering module. The training set of RR-Net has 30,000 images from multiple predefined views and it’s good at generalizing on novel views. We set the learning rate as 2×10^{-4} and batch size as 16 for 800 epochs.

Results. Depth estimation, raycasting, and color blending are performed using parallel computations via CUDA operators. The PyTorch model of RR-Net is converted to TensorRT, where the input and output processes also leverage CUDA kernels. The time consumption for each part on NVIDIA RTX 4080 is detailed in Table 1. PSNR of color blended and RR-Net results are 24.84 and 25.20, respectively. From the user’s perspective, the rendered results can be aligned with the specular reflections on the 3D display.

4 CONCLUSION

In this paper, we propose an AvatarMirror framework with instant novel-view 3D reconstruction and neural rendering. Evaluation results show that our framework achieves real-time and high-fidelity digital human presence, boosting the immersive sense. The proposed VR framework is promising for use in telepresence and immersive virtual conferences.

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